Roll No :		**Name :**	
Experiment No :	**5**		
Title :	Simulate BER performance of multi-antenna Rayleigh channel for SNR varying from 0 to 60 dB.		
Date of Performance :			
Date of Checking :			
Grade :		**Sign of Staff**	

EXPERIMENT NO. 5
BER PERFORMANCE OF
MULTI-ANTENNA RAYLEIGH CHANNEL

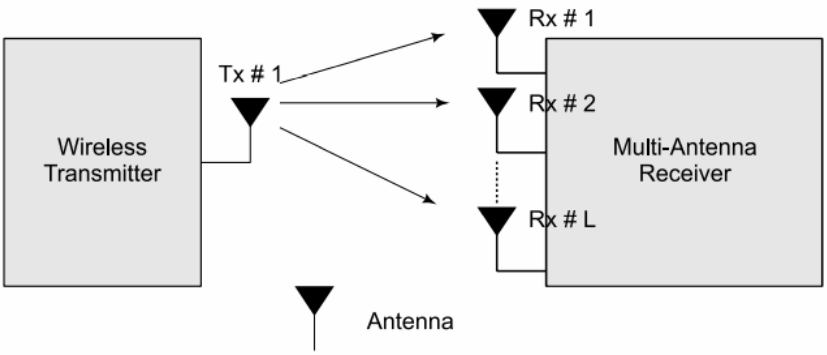
TITLE:- Simulate BER performance of multi-antenna Rayleigh channel for SNR varying from 0 to 60 dB.

OBJECTIVE OF EXPERIMENT:- Simulate BER performance of multi-antenna Rayleigh channel for SNR varying from 0 to 60 dB.

APPARATUS:- Matlab, PC, Printer

THEORY:-

Multiple Receive Antenna System Model We start the diversity analysis of the multi-antenna system by initially developing an analytical model for the system illustrated in Figure



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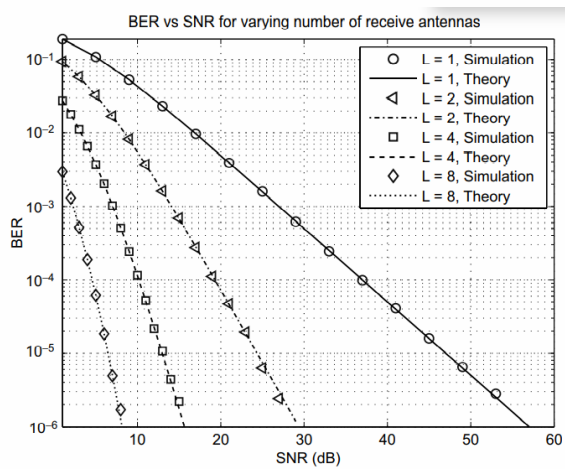
Consider the transmission of a BPSK data symbol x (k) at the time instant k from the single transmit antenna, i.e., Tx #1. If each symbol is of average power P, it naturally follows that x (k) = ± √ P, where a transmission of + √ P , − √ P corresponds to the information bits 1, 0 respectively.

Since there are L receive antennas, there is correspondingly a wireless channel between the transmit antenna and each receive antenna, i.e., L channels in total over which the signal is received at the receiver. Let the channel coefficient between the transmit antenna and the ith receive antenna be denoted by hi, which is the complex Rayleigh fading coefficient.



Since the exact computation of the above expression for BER is quite complex, we will skip it here. The final expression for the BER of this multi-antenna wireless system with MRC combining is given as

$$\mathrm{BER}_{\text{Multi-Antenna}} = \left(\frac{1-\lambda}{2}\right)^{L} \sum_{l=0}^{L-1} {}^{L+l-1}C_l \left(\frac{1+\lambda}{2}\right)^{l} \tag{3.17}$$

where the quantity λ is defined as $\lambda = \sqrt{\frac{\mathrm{SNR}}{2+\mathrm{SNR}}}$, with SNR denoting the average receiver SNR for each link given by $\frac{P}{\sigma_n^2}$ and ${}^{n}C_k = \frac{n!}{k!(n-k)!}$. Notice that for $L = 1$, i.e., the case with a single receive antenna, the above expression reduces to

$$\mathrm{BER} = \left(\frac{1-\lambda}{2}\right)^{1} \sum_{l=0}^{0} {}^{0}C_l \left(\frac{1+\lambda}{2}\right)^{l} = \frac{1}{2}(1-\lambda)$$

CODE:-

OBSERVATION TABLE:-

With Two Antenna Receiver L=2

Sr. No.	Number of Antennas L	SNR dB	SNR ($Y.dB=10^{Y/10}$)	$\lambda = \sqrt{\frac{SNR}{2+SNR}}$	BER
1	2	0			
2	2	10			
3	2	20			
4	2	30			

With Four Antenna Receiver L=4

Sr. No.	Number of Antennas L	SNR dB	SNR ($Y.dB=10^{Y/10}$)	$\lambda = \sqrt{\frac{SNR}{2+SNR}}$	BER
1	4	0			
2	4	2			
3	4	5			
4	4	10			

RESULTS:-

Graph work:

1. Plot graph of SNR in dB versus BER with L = 2
2. Plot graph of SNR in dB versus BER with L = 4

CONCLUSION:-

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